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Monitoring of swarming sounds in bee hives for early detection of the swarming period

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Abstract

Beekeeping, known as one of the oldest forms of agriculture, in its complexity requires control for honey production with what modern technology can offer. Honey is included in animal production implying that farmers have interest in big productions according to the best blooming time, the presence of parasites, the genetic strain of his bees and the swarming periods of the honeybees (queen and her workers leaving the hive).

This last fact has a big economic interest for the beekeeper as swarming means honey loss since bees start collecting the honey to migrate. Here for a method that enables the prediction of the swarming is required to prevent the queen from leaving the hives. In this experiment an acoustic method based on labelling of sounds is proposed to predict the swarming period. Three hives were monitored during 270 h. The microphones were sited inside the hives together with a temperature and humidity sensor. The sounds were recorded with a sample rate of 2 kHz, and analyzed via Matlab and Cool Edit Pro. During this period 9 swarming activities occurred. Swarming is indicated by an increase in the power spectral density at about 110 Hz; approaching to swarm the sound augmented in amplitude and frequency to 300 Hz, occasionally a rapid change occurred from 150 Hz to 500 Hz. Another finding indicating the initiation of a swarming period is the raise in temperature from 33 °C to 35 °C until the actual time of swarming when the temperature starts dropping to 32 °C. With more activity, ventilation from bee wings causes drop of temperature. Less information comes from the correlation between sound and humidity since this parameter is too much influenced by the external conditions and no significant variation occurred according to a swarm. This increase of temperature, together with the changes in acoustical features of the sound recorded in the hive, may be used as a predictor for swarming of the bees to reduce honey loss.

Keywords

Bee; Frequency; Hive; Honey loss; Sounds; Swarming; Temperature

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Introduction

Beekeeping is known to be one of the oldest forms of agriculture: Publius Vergilius Maro (Mantova, 15 October 70 a.C.—Brindisi, 21 September 19 a.C.) in his Georgicon wrote about the art of beekeeping, therefore this discipline has progressed gradually together with human agricultural knowledge and technologies and its complexity implies modern technology control for a bigger honey production. Beekeeping is agriculture sector with rural development functions, for honey and hive's goods production and the maintenance of ecological balance. To improve production and marketing of apiculture products, program protocols should be drawn up comprising technical assistance, control of varroassis, rationalization of transhumance, management of hive restocking in the Community and co-operation on research programs between partners.

Honey is an animal production and the farmer has interest in big productions according to the best blooming time, the presence of parasites, the genetic strain and the swarming periods of the honeybees.

Swarming represents the natural way a honey bee colony uses to reproduce. Swarming normally occurs in strong populous colonies and chiefly from spring to early summer. The preparations for lift-off takes almost 1 month (Lindauer, 1955). About 10 days prior to leaving the hive, the bee workers engorge themselves honey (36 mg for each bee) and nearly cease normal flight activity; few hours before swarming, some workers become ‘excited’ and “begin running back and forth in waves, buzzing to excite the other workers” (Winston, 1987). This last fact has a big economic and managerial interest for the beekeeper as swarming means bee and honey loss.

Three factors that are connected with swarming are the brood and adult bee population (as related to space within the hive), the amount and distribution of available queen pheromone and the hive environment. All these factors are related to be a combination of sounds, temperature and humidity of the hive system itself. Beekeepers must take responsibility for their swarms since control of swarming involves more extensive manipulation and rather critical timing. Several scientists widely worked to predict swarming moments by individuating the main factors that could lead a hive to swarm. Early studies attribute the cause of swarming to the amount of food in the hive. Morland (1930) argued that swarming was the result of too many young nurse bees that produced too much brood food. In order to reduce the amount of food, queen rearing began, which led to swarming. Another group of scientists believed that the crowding of adult workers and limited space for food rearing resulted in swarming (Demuth, 1921, Winston, 1987). There is, however, no sufficient evidence to support these hypotheses (Winston, 1987). Other investigations (Simpson, 1958, Winston et al., 1980, Lensky and Slabezki, 1981) found a multifactor relationship between swarming and several within-colony demographic factors such as colony size, brood hive congestion, worker age distribution and reduced transmission of queen substances. Seeley and Heinrich (1980) considered the temperature of the system before lift-off. Camazine and Visscher (1999) and Lewis and Schneider (2000) studied their communication by vibration signals and dances and other researchers considered sound (Woods, 1959; Vancata, 1995; Wenner, 1962, Esch, 1967). None of these factors alone are found to initiate swarming (Winston, 1987). This analysis will consider the anamnesis of the hives, the sound, the external and internal temperature and relative humidity.

By understanding how to best predict swarming, beekeeping will be improved, more enjoyable and profitable.

A method that enables the prediction of the swarming is proposed to prevent the queen from leaving the hives by using sound analysis.

By referring to sound data collected in hives during the swarming seasons we can study the sound features correlated with the activity of the insects, temperature and humidity of the hive ecosystem.

The aim of the study will to build up an early swarming monitoring system that will inform the beekeeper about the trends of his hives and manage them only in critical moments when necessary, avoiding loss of honey.

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Materials and methods

In this experiment an acoustical method based on sound analysis for classification is proposed to identify the swarming period of the Italian Honey Bee *Apis Mellifera Ligustica*. Three hives were monitored with sound, humidity and temperature during 270 h continuous recording in May 2006 in the beekeeping experimental farm from the University of Milan (Fig. 1). The beekeeper did not remove the newborn queen cells to allow as much as possible recording of swarms sounds; this practice is in reality …

Results

Nine swarming activities occurred during the recordings which were detected both from sound and visual analysis of the observer in situ at the right moment.

Swarms occurred always in daylight time and preferentially in the hottest hours. The duration of a swarm (moment from which the bees started to excite until they leave the hive) cannot be considered as a standard parameter to discriminate the sounds from the others since they had wide range in length lasting from 13.53 min to 56.23 min (μ=35 …

Conclusions and discussion

In this project 270 h of sound recording was done in bee hives together with the recordings of temperature and humidity. The aim was to investigate the changes in sound characteristics and the relation with changes in temperature and humidity. It could be seen that with increase of activity, as seen at swarming moments, the frequency content shifts from about the range of 100–300 Hz to a higher frequency range of 500 Hz to 600 Hz. This is caused by the intense flitting of the wings, which causes …

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> …In this sense, precision beekeeping becomes a handy tool for the study of bee colonies. For instance, the application of electronic systems and communication developments is becoming more and more frequent to reach this objective (Dunham, 1931; Woods, 1958; Kronenberg and Heller, 1982; Dietlein, 1985; Fahrenholz et al., 1989; Reynolds and Riley, 2002; Vornicu and Olah, 2004; Human et al., 2006; Ferrari et al., 2008; Atauri and Llorente, 2009; Meitalovs et al., 2009; Bencsik et al., 2011; Eskov and Toboev, 2011; Rice, 2013; Stalidzans and Berzonis, 2013; Zacepins and Karasha, 2013; Meikle and Holst, 2015; Murphy et al., 2015; Sánchez et al., 2015; Zacepins et al., 2015; Gil-Lebrero et al., 2017). The possibility of applying commercial electronic systems, known as data loggers, is the base of precision beekeeping.…

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